

Derivative-free low-chaos reconstruction and the adapted Cartan

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

first issued 2026-07-29 · this version issued 2026-07-29 · v1.0

1. Purpose and scope

R-121 reduced the surviving first-order payment to an adapted fifth $H^{-3/5}$ coefficient moment, but left two questions entangled: whether the actual zero/first Wiener defects require differentiating the feedback, and whether source energy plus the terminal sextic automatically supplies that fifth moment. The present five-round audit separates them.

R-122 conclusion. At fixed cutoff and positive floor, the actual endpoint defects D_0, D_1 have exact derivative-free formulas in terms of the joint law of the complete endpoint output and its once-owned trace. No Dh_2 , D^2h_2 , Faà di Bruno forest, or termwise adjacent-chaos expansion is needed to identify them. This is a real reconstruction advance, not a proof that they vanish. The R-075 graph topology does not control feedback derivatives. More decisively, an exact coherent-amplitude family has uniformly bounded source L^2 and terminal L^6 budgets while its quadratic $H^{-3/5}$ current has divergent fifth moment. The surviving rational A1 Cartan coefficient has a nonzero quadratic coherent ray. Thus a standalone adapted fifth-moment inference from the existing two budgets is false. Formal selfadjointness of the complete endpoint Hessian does not cancel the first-order Cartan coefficient. On the normalized R-102 slice, the Cartan piece $1360J/729$ and square-cross piece $1320J/729$ reinforce to $2680J/729$. The surviving route is derivative-free and correlation-preserving: retain the current, score, trace, low-chaos, and R-063 forest owners in one signed expectation before applying Young. R-122 does not prove that complete estimate. Both A13 gates, $\text{OVERLAP}_{\text{src}}$, Nelson, removals, the interacting measure, and Sector-A closure remain open. Tier stays T4.

The five rounds are: endpoint reconstruction, identifiability/topology, adapted fifth moment, complete Euler–Lagrange cancellation, and synthesis of the shortest surviving route. Failed methods are recorded because they remove recurrent ambiguity rather than because they refute A13.

2. Round one: the actual derivative-free endpoint kernels

Use the legal R-119 two-root/two-visit chart. Let $\xi = \xi_1$ and $\eta = \xi_2$ be independent fresh Gaussian blocks and write

$$Z_* = S_1(\xi + h_1) + S_2(\eta + h_2(\xi)). \quad (2.1)$$

For one root and one contraction-closed output cluster, define the complete conditional endpoint current

$$\Phi_h(\xi)(r) = \Pi_C \mathbb{E}_\eta[\Xi_\Sigma(Z_*, \partial Z_*)(r) \mid \xi]. \quad (2.2)$$

The heat variable r remains a coordinate in the output Hilbert space; it is not averaged away in (2.2). Put

$$b = \mathbb{E}\Phi_h, \quad Y = \Phi_h - b, \quad A = \mathbb{E}[Y \otimes \xi], \quad R = Y - A\xi. \quad (2.3)$$

Then R is orthogonal to zero and first chaos.

Let $\Theta_h(\xi)$ denote the once-owned complete trace. In the R-119 notation it is the live $P_1\tau_{*,1,C}$ coordinate,

$$\Theta_h(\xi) = \mathbb{E}_\eta \left[\sum_{i,n \in \mathcal{C}} \int \mathbb{E}_{G_1} \left| C_* \widehat{(\cdot, r)} \partial_i G_1(n) \right|^2 \mu_\Sigma(dr) \middle| \xi \right]. \quad (2.4)$$

This is the safe trace owner. Its R-063 forest is an alternative expansion, not an extra term.

Theorem 2.1 (derivative-free D_0, D_1 reconstruction)

Assume the displayed moments are finite and put $H_2(\xi) = \xi \otimes \xi - \mathbf{I}$. Then

$$r_2 = \frac{1}{2} \mathbb{E}[R \otimes H_2(\xi)], \quad t_0 = \mathbb{E}\Theta_h - \|A\|_{\text{HS}}^2, \quad t_1 = \mathbb{E}[\xi \Theta_h], \quad (2.5)$$

and the R-119 aggregate low-chaos defects are

$$\boxed{D_0 = \mathbb{E}\Theta_h - \|A\|_{\text{HS}}^2 - \mathbb{E}\|R\|^2}, \quad (2.6)$$

$$\boxed{D_1 = \mathbb{E}[\xi \Theta_h] - 2 \text{Contr}(A, \mathbb{E}[R \otimes H_2(\xi)]) - \mathbb{E}[\xi \|R\|^2]}. \quad (2.7)$$

Here $(\text{Contr}(A, T))_i = \sum_j \langle Ae_j, T(i, j) \rangle$.

Proof. At finite cutoff, Gaussian integration by parts gives

$$\mathbb{E}[R \otimes H_2(\xi)] = \mathbb{E} D^2 R, \quad qquad \mathbb{E}[\xi \|R\|^2] = 2 \mathbb{E} \langle D_i R, R \rangle. \quad (2.8)$$

Consequently

$$\sum_{n \geq 3} n! \langle r_{n-1}, r_n(\cdot, i) \rangle = \frac{1}{2} \mathbb{E}[\xi_i \|R\|^2]. \quad (2.9)$$

Insert (2.8)–(2.9) into R-119 (3.9)–(3.10), using $\delta\tau = \Theta_h - \|A\|_{\text{HS}}^2$. This proves (2.5)–(2.7). $\square$

The right sides of (2.6)–(2.7) use only the endpoint law. Gaussian integration by parts proves the identity, but evaluating it does not require differentiating the feedback map. Passing to a limiting graph still needs uniform integrability of the listed products.

3. Round two: identifiability and graph topology

The derivative-free representation prevents a false inference in the other direction: chart geometry and source cost do not determine its values. Consider the normalized circle basis

$$e_0 = 1, \quad e_s = \sqrt{2} \sin x, \quad e_c = \sqrt{2} \cos x, \quad e_{s2} = \sqrt{2} \sin 2x,$$

and the legal bounded first-linear-row fixture

$$u_h = \xi e_s + (\eta + h(\xi)) e_0, \quad qquad J_h = \kappa u_h \partial_x u_h, \quad qquad \kappa = 2\sqrt{c_0}. \quad (3.1)$$

Its complete row cluster is

$$Y_h = \kappa \left[(\xi h - \mathbb{E}\xi h) e_c + \frac{H_2(\xi)}{\sqrt{2}} e_{s2} \right], \quad \Theta_h = \kappa^2 \left(\frac{\xi^2}{2} + h^2 + 1 \right). \quad (3.2)$$

Take

$$h_\pm(\xi) = a \sin \xi \pm d \cos \xi. \quad (3.3)$$

Both controls are bounded, smooth, and strict-past. They have the same cost

$$\mathbb{E}h_{\pm}^2 = \frac{a^2(1 - e^{-2})}{2} + \frac{d^2(1 + e^{-2})}{2}, \quad (3.4)$$

the same $A = 0$, r_2 , t_0 , and D_0 , with

$$r_2 = \kappa \left[\frac{ae^{-1/2}}{2} e_c + \frac{1}{\sqrt{2}} e_{s2} \right], \quad (3.5)$$

$$\frac{D_0}{\kappa^2} = \frac{1}{2} + a^2(e^{-1} - 2e^{-2}) + 2d^2e^{-2}. \quad (3.6)$$

Nevertheless

$$t_{1,\pm} = \pm 2\kappa^2 ade^{-2}, \quad \frac{1}{2} \mathbb{E}[\xi \|R_{\pm}\|^2] = \mp \kappa^2 ade^{-2}, \quad (3.7)$$

and hence

$$D_{1,\pm} = \pm 4\kappa^2 ade^{-2}. \quad (3.8)$$

Thus even the source cost together with A, r_2, t_0, D_0 does not identify the first-chaos trace, adjacent-chaos contraction, or D_1 .

There is also a topology obstruction to the old derivative-by-derivative route. For $\xi \sim N(0, 1)$ set

$$h_n(\xi) = \frac{\sin(n\xi)}{n}. \quad (3.9)$$

Then

$$\mathbb{E}h_n^2 = \frac{1 - e^{-2n^2}}{2n^2} \longrightarrow 0, \quad (3.10)$$

$$\mathbb{E}|h_n|^6 = \frac{10 - 15e^{-2n^2} + 6e^{-8n^2} - e^{-18n^2}}{32n^6} \longrightarrow 0, \quad (3.11)$$

whereas

$$\mathbb{E}|Dh_n|^2 = \frac{1 + e^{-2n^2}}{2} \longrightarrow \frac{1}{2}, \quad \mathbb{E}|D^2h_n|^2 = \frac{n^2(1 - e^{-2n^2})}{2} \longrightarrow \infty. \quad (3.12)$$

Therefore R-075 graph convergence cannot justify separate limits of the four Dh_2, D^2h_2 chain-rule families. This is a method no-go, not a failure of (2.6)–(2.7); the law formulas are precisely the repair.

4. Round three: the exact adapted fifth-moment boundary

R-071 gives, for an unshifted stationary Gaussian field and common fixed floor, every finite $L^p(\Omega; H^{-1/2-\delta})$ moment of its smooth one-form current. Taking $\delta = 1/10$ includes $L^5(H^{-3/5})$. R-063 permits a deterministic Cameron–Martin translation with its complete forest retained, but gives no uniform random-shift integrability. R-075 passes a graph limit only after the adapted current bound and identification are already supplied.

Proposition 4.1 (a valid conditional shifted theorem)

Let U be a fresh R-071 Gaussian field, let h be measurable with respect to a past sigma-field $\mathcal{G}$, and assume that U is conditionally independent of $\mathcal{G}$. For every $r > 1$,

$$\mathbb{E}[\|J(U + h)\|_{H^{-3/5}}^5 \mid \mathcal{G}] \leq \|J(U)\|_{L^{5r}(H^{-3/5})}^5 \exp\left(\frac{\|h\|_{\text{CM}}^2}{2(r-1)}\right). \quad (4.1)$$

Proof. Apply Hölder to the Cameron–Martin density $L_h = \exp(\langle h, U \rangle - \|h\|_{\text{CM}}^2/2)$ with conjugate exponent $r' = r/(r-1)$. Its required norm is

$$(\mathbb{E} L_h^{r'})^{1/r'} = \exp\left(\frac{\|h\|_{\text{CM}}^2}{2(r-1)}\right). \quad \square$$

Thus a positive exponential moment of $\|h\|_{\text{CM}}^2$ is sufficient. The R-093 progressive/revisit union supplies only quadratic source energy and does not meet this hypothesis; same-root later feedback is not conditionally fresh.

For an owner-complete fresh forest of chaos order at most four, the $p = 5$ hypercontractive factors are $4^{3/2} = 8$ and $4^2 = 16$. Hilbert BDG then reduces the fifth moment to the missing bracket datum

$$\mathbb{E} \left(\sum_k \mathbb{E}_{k-1} \|D_k\|_{H^{-3/5}}^2 \right)^{5/2} < \infty, \quad (4.2)$$

up to the BDG constant and the worst fifth-power factor $16^5 = 2^{20}$. Neither (4.2) nor an equivalent tenth coherent-amplitude bound is presently available.

Theorem 4.2 (coherent-amplitude graph nontransfer)

On normalized $\mathbb{T}^3$, let $\xi \sim N(0, 1)$ and

$$A_t = e^{t\xi - 3t^2}, \quad z_t = A_t(\cos x_1, \sin x_1), \quad J(v) = v_1 \partial_1 v_2. \quad (4.3)$$

For every $p > 0$,

$$\mathbb{E} A_t^p = e^{p(p-6)t^2/2}. \quad (4.4)$$

Consequently

$$\mathbb{E} \|z_t\|_{H^2}^2 = 3e^{-4t^2} \longrightarrow 0, \quad \mathbb{E} \|z_t\|_6^6 = 1, \quad (4.5)$$

but, since $J(z_t) = A_t^2 \cos^2 x_1$,

$$\mathbb{E} \|J(z_t)\|_{H^{-3/5}}^5 = c_J^5 e^{20t^2} \longrightarrow \infty, \quad (4.6)$$

where

$$c_J^2 = \|\cos^2 x_1\|_{H^{-3/5}}^2 = \frac{1}{4} + \frac{1}{8} 5^{-3/5} > 0. \quad (4.7)$$

Smooth caps $0 \leq \chi_N(s) \leq s$ can be chosen to agree with s below N . For each t , choose $N(t)$ large enough to retain at least one half of the tenth moment. The capped amplitudes are bounded smooth Gaussian cylinders, their second and sixth moments do not increase, and (4.6) still diverges along a sequence. Hence existing graph budgets do not imply the standalone fifth current moment even on bounded cylindrical cores. $\square$

5. The production rational coefficient has the same quadratic ray

The preceding theorem is not merely an arbitrary scalar integrability trick. Use the active A1 doublet generator $S = \text{diag}(1, -1)$ and the profile

$$\phi(x) = (2 + \cos x, \sin x), \quad \rho_\phi = 5 + 4 \cos x, \quad m_\phi = 4 + 4 \cos x + \cos 2x. \quad (5.1)$$

For $u_H = H\phi$, the R-120 rational one-form is

$$\theta = dm - \alpha \frac{m}{\rho + e} d\rho, \quad \alpha = \frac{5}{9}. \quad (5.2)$$

Write $j_H = \theta(u_H)(\partial_x u_H)$ and let ω_H be the scalar coefficient of its curvature. Directly,

$$\frac{j_H}{H^2} \longrightarrow m'_\phi - \alpha \frac{m_\phi}{\rho_\phi} \rho'_\phi, \quad \omega_H \longrightarrow -\frac{8\alpha(2 + \cos x) \sin x}{\rho_\phi}. \quad (5.3)$$

At $x = \pi/2$ these limits are

$$-\frac{8}{3}, \quad -\frac{16}{9}, \quad \boxed{\lim_{H \rightarrow \infty} \frac{j_H \omega_H}{H^2} = \frac{128}{27} \neq 0}. \quad (5.4)$$

Because $\rho_\phi \geq 1$, convergence holds smoothly and the limiting coefficient is a nonzero $H^{-3/5}$ function. Therefore the corresponding production Cartan subcoefficient grows quadratically along this coherent ray, and the capped lognormal construction applies to a coefficient-first fifth-moment estimate.

This does not refute the complete signed A1 Hessian. The $D_z \Omega$ term, square, trace, low-chaos, and R-063 forest owners can remain correlated and can cancel or be paid jointly. It refutes only the black-box inference that their isolated adapted coefficient automatically lies in $L^5(H^{-3/5})$ from the two existing graph budgets.

6. Round four: complete Euler–Lagrange selfadjointness

The remaining proposed shortcut was that the complete scalar endpoint Hessian might cancel the first-order Cartan block automatically. Let

$$\mathcal{E}[U] = \frac{1}{2} \sum_i \int B(U) : (G_i \otimes G_i - \Gamma_i), \quad G_i = \partial_i U, \quad (6.1)$$

with $B = B^T$, and define

$$Q_i = G_i \otimes G_i - \Gamma_i, \quad qquad(C_i)_{ac} = \partial_c B_{ab} G_i^b. \quad (6.2)$$

The unintegrated symmetric Hessian is

$$H_U(h, k) = \sum_i \int \{ (\partial_i h)^T B \partial_i k + (\partial_i k)^T C_i h + (\partial_i h)^T C_i k + \frac{1}{2} D^2 B[h, k] : Q_i \}. \quad (6.3)$$

After integration by parts its Jacobi operator is

$$\mathcal{L}_U h = - \sum_i \partial_i (B \partial_i h) + \sum_i (A_i \partial_i h + V_i h), \quad (6.4)$$

where

$$A_i = C_i^T - C_i, \quad qquad(A_i)_{dc} = (\partial_d B_{cb} - \partial_c B_{db}) G_i^b. \quad (6.5)$$

The exact adjoint relation is

$$A_i^T = -A_i, \quad qquad V_i = V_i^T = \partial_i A_i. \quad (6.6)$$

Thus

$$\boxed{\mathcal{L}_U = -\partial_i (B \partial_i) + A_i \partial_i + \frac{1}{2} \partial_i A_i + S_i, \quad qquad S_i = S_i^T}. \quad (6.7)$$

Selfadjointness is achieved by the skew first-order completion, not by $A_i = 0$. The covariance trace Γ_i appears only in the symmetric zeroth-order term and cannot alter A_i .

Theorem 6.1 (exact normalized R-102 noncancellation)

On the R-102 active slice use

$$g = \left(x - \frac{5}{9} \frac{x^3}{1+x^2+y^2}, -\frac{5}{9} \frac{x^2 y}{1+x^2+y^2} \right), \quad \text{quad} B = 4gg^T, \quad \text{quad} J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad (6.8)$$

At $u_2 = (2, 1)$ and $G = e_1$,

$$A_2 = \frac{2680}{729} J. \quad (6.9)$$

Factor $B = f f^T$ with $f = 2g$. Then

$$f = \frac{1}{27}(68, -20), \quad \ell = (Df)^T G = \frac{1}{81}(62, 40), \quad \Omega = \frac{20}{27} J, \quad (6.10)$$

and the full first-order coefficient splits as

$$A_2 = j\Omega + \ell \otimes f - f \otimes \ell = \frac{1360}{729} J + \frac{1320}{729} J = \frac{2680}{729} J. \quad (6.11)$$

The two contributions reinforce. At $u_0 = (1, 1)$,

$$A_0 = \frac{1480}{729} J, \quad \text{quad} A_2 - A_0 = \frac{400}{243} J \neq 0. \quad (6.12)$$

Thus even co-varying both endpoints does not cancel the coefficient; in the actual fixed- U_0 problem all of A_2 remains.

As a convention-independent check, take $\theta = (-y/2) dx + (x/2) dy$ and $u(t) = (\cos t, \sin t)$ on $[0, 2\pi]$. Along the radial variation $z = u$,

$$\mathcal{E}((1+s)u) = \frac{\pi}{4}(1+s)^4, \quad \text{quad} \mathcal{E}''(0) = 3\pi = 2\pi + \pi, \quad (6.13)$$

where 2π is the square term and π is the Cartan term. The latter is nonzero and necessary for the symmetric Hessian.

The theorem rules out cancellation inferred solely from scalar exactness, endpoint telescoping, formal selfadjointness, torus integration, or the covariance trace. It does not rule out a separately proved identity after the full control projection or an expectation-level owner cancellation.

7. Round five: correlation-preserving boundary

The failure of the isolated fifth moment reveals where the old proof spends too much. Along a coherent amplitude ray, the current coefficient and its fixed-skew payload both scale as H^2 . Their joint signed pairing therefore scales as H^4 , not as the tenth power introduced by taking a separate fifth coefficient moment. For $x = H^2$ and $c, \zeta > 0$,

$$\boxed{cx^2 \leq \zeta x^3 + \frac{4c^3}{27\zeta^2}.} \quad (7.1)$$

Indeed the maximum of $cx^2 - \zeta x^3$ occurs at $x = 2c/(3\zeta)$. Hence the terminal sextic can pay the correlated coherent amplitude after the coefficient and payload have been multiplied.

Equation (7.1) is not the full A13 theorem. It is a sharp route boundary: separating the coefficient before Young can demand a false tenth-amplitude datum, whereas retaining the complete signed owner can remain within the existing sextic scale. The next target is therefore a derivative-free form estimate of the schematic type

$$\left| \mathbb{E} \int \mathcal{P}_{\text{complete}}(h) \right| \leq \eta \mathbb{E} \|L_\pi h\|_{H^2}^2 + \zeta \mathbb{E} \|Z_h\|_6^6 + C_{\eta, \zeta}, \quad (7.2)$$

where $\mathcal{P}_{\text{complete}}$ contains the current, square, trace, finite-low, score, and R-063 forest owners before absolute values. It must be uniform in cutoff, chart, visit, and revisit and must spend the R-093 directed source union and terminal sextic only once.

Two alternatives remain logically valid but require genuinely new data: the conditional Cameron–Martin exponential moment in (4.1), or the $L^{5/2}$ predictable-bracket estimate in (4.2). Neither follows from the current authorities.

8. Devil’s-advocate

1. **Objection:** (2.6)–(2.7) prove $D_0 = D_1 = 0$. **Disposition:** **DISMISSED**. They reconstruct the defects. The bounded fixture (3.3)–(3.8) gives opposite nonzero D_1 with identical lower data.
2. **Objection:** the h_n sequence refutes endpoint graph recovery. **Disposition:** **DISMISSED**. It refutes only recovery of feedback derivatives from the R-075 graph topology. The derivative-free law route survives, subject to uniform integrability.
3. **Objection:** Theorem 4.2 is an A1 or one-use counterexample. **Disposition:** **DISMISSED**. It is a counterexample to a separated coefficient-moment inference. Complete signed owners may remain correlated.
4. **Objection:** the coherent amplitude is unbounded and therefore illegal. **Disposition:** **DISMISSED**. Smooth bounded caps preserve uniform second/sixth budgets and an unbounded subsequence of tenth moments.
5. **Objection:** formal selfadjointness forces the first-order coefficient to vanish. **Disposition:** **DISMISSED**. Equations (6.6)–(6.7) show the exact skew completion, and (6.11) is nonzero.
6. **Objection:** the covariance trace can cancel A_i . **Disposition:** **DISMISSED**. It enters only the symmetric zeroth-order Hessian. The independent rational-jet executable verifies this separation.
7. **Objection:** (7.1) closes the production one-use theorem. **Disposition:** **UPHELD AS A BOUNDARY**. It closes only the coherent amplitude scaling test. Spatial multipliers, signs, owner completeness, same-root Jensen defects, and uniform aggregation remain to be proved.
8. **Objection:** exact fractions were copied as conclusions. **Disposition:** **DISMISSED**. The primary route derives α from the A1 manifest and differentiates the rational coefficient symbolically. The non-importing route uses an independent exact rational first-jet algebra. The displayed fractions are explicit test oracles in both audits.

9. Evidence map and next checkpoint

The successful chain is

$$\text{R-119 endpoint law} \implies \text{derivative-free (2.6)–(2.7)} \implies \text{complete-law evaluation target.} \quad (9.1)$$

The failed chains are

$$\text{R-075 graph} \not\Rightarrow Dh, D^2h, \quad (9.2)$$

$$L^2 \text{ source} + L^6 \text{ terminal} \not\Rightarrow L^5(H^{-3/5}) \text{ isolated current}, \quad (9.3)$$

$$\text{scalar Hessian selfadjointness} \not\Rightarrow \text{Cartan cancellation.} \quad (9.4)$$

The surviving arrow is

$$\text{complete derivative-free signed owner} \implies \text{correlation-preserving form estimate (7.2),} \quad (9.5)$$

and (9.5) is a target, not yet a theorem.

The next checkpoint must construct Φ_h, Θ_h after summing all six A1 rows and all required current/trace/low/rational/R-063 companions, then evaluate (2.6)–(2.7) and estimate the complete form without isolating a fifth coefficient moment. A failure must identify the precise surviving signed defect, not reopen any of (9.2)–(9.4).

10. Result footer

The primary executable checks 82 exact symbolic assertions: law/derivative equivalence, the bounded owner fixture, graph topology, coherent amplitude, conditional sufficient conditions, the production rational ray, complete R-102 Cartan reinforcement, and the optimal joint Young constant. The non-importing standard-library executable checks 57 assertions using exact Gaussian polynomials, exact rational first jets, independent characteristic arithmetic, and separate scope gates. The integrated verifier reruns both, pins all authorities and hashes, checks the paired PDF and public surfaces, and enforces every no-overclaim boundary.

R-122 proves no production D_0, D_1 cancellation, uniform complete signed one-use form, $\text{OVERLAP}_{\text{src}}$, Nelson estimate, cutoff or floor removal, interacting measure, Sector-A closure, or T5–T7 promotion.

Result ID: A13-CLASSII-DERIVATIVE-FREE-LOW-CHAOS-ADAPTED-FIFTH-MOMENT-CARTAN-BOUNDARY (R-122).
Precise statement: Thm. 2.1; Sec. 3; Props. 4.1 and Thm. 4.2;
Eqs. (5.1)–(5.4); Thm. 6.1; Eq. (7.1).
Scope: fixed finite cutoff and positive rational floor; conditional fresh-shift theorem only; full progressive/revisit estimate open.
Dependencies: A1, R-063, R-071, R-075, R-102, R-119, R-120, R-121.
Evidence grade: ANALYTIC, EXACT, EXECUTED.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/a13_classii_derivative_free_low_chaos_adapted_fifth_moment_cartan_boundary_verify.py
Expected output: R-122 PASS 122/122; primary
82/82; independent 57/57; all hash, PDF, route, registry, and no-overclaim gates pass.
Falsification gate: any failed low-chaos, owner-fixture, graph, moment, rational-ray, Cartan, Young, authority, PDF, or surface check.
Tier before / after: T4 / T4.
No-overclaim statement: production cancellation, complete one-use, $\text{OVERLAP}_{\text{src}}$, Nelson, removals, measure, and Sector A remain open.
Next required action: evaluate the complete derivative-free endpoint law and prove the correlation-preserving signed owner estimate once.